



Brain tumor detection and segmentation using deep learning

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Abstract

Objectives Brain tumor detection, classification and segmentation are challenging due to the heterogeneous nature of brain tumors. Different deep learning-based algorithms are available for object detection; however, the performance of detection algorithms on brain tumor data has not been widely explored. Therefore, we aim to compare different object detection algorithms (Faster R-CNN, YOLO & SSD) for brain tumor detection on MRI data. Furthermore, the best-performing detection network is paired with a 2D U-Net for pixel-wise segmentation of abnormal tumor cells.

Materials and methods The proposed model was evaluated on the Brain Tumor Figshare (BTF) dataset, and the best-performing detection network cascaded with 2D U-Net for pixel-wise segmentation of tumors. The best-performing detection network was also fine-tuned on BRATS 2018 data to detect and classify the glioma tumor.

Results For the detection of three tumor types, YOLOv5 achieved the highest mAP of 89.5% on test data compared to other networks. For segmentation, YOLOv5 combined with 2D U-Net achieved a higher DSC compared to the 2D U-Net alone (DSC: YOLOv5 + 2D U-Net = 88.1%; 2D U-Net = 80.5%). The proposed method was compared with the existing detection and segmentation network i.e. Mask R-CNN and achieved a higher mAP (YOLOv5 + 2D U-Net = 89.5%; Mask R-CNN = 67%) and DSC (YOLOv5 + 2D U-Net = 88.1%; Mask R-CNN = 44.2%).

Conclusion In this work, we propose a deep-learning-based method for multi-class tumor detection, classification and segmentation that combines YOLOv5 with 2D U-Net. The results show that the proposed method not only detects different types of brain tumors accurately but also delineates the tumor region precisely within the detected bounding box.

Keywords Magnetic resonance imaging (MRI) · Brain tumor · Deep learning · Segmentation · Detection · Classification

Introduction

Brain tumors develop from the abnormal and uncontrolled division of cells in the brain. Brain tumors can be categorized as primary or secondary brain tumors. The three prominent types of primary brain tumors are meningioma, glioma and pituitary tumor which account for the highest incidence

rates among all the primary brain tumors [1]. The abnormal tumor cells in the brain are identified by scanning the brain regions using different medical imaging techniques. In current medical scanning techniques, Magnetic Resonance Imaging (MRI) is superior to Computed Tomography (CT) due to its high sensitivity and better contrast with respect to various intensity tissues in brain images [2].

Accurate detection and segmentation of brain tumor is a challenging task due to its heterogeneous nature and ambiguous boundaries between the normal and malignant tissues [3]. Manual detection of tumor can result in accurate prediction and diagnosis, but it becomes a tedious task when there is a large amount of data present [4]. Therefore, it is important to develop a highly accurate and automated brain tumor detection and segmentation method.

Object detection is one of the extensively researched topics in the field of computer vision. The aim of localizing the object in an image can be achieved by defining bounding box coordinates (longitude and latitude) around a target

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object. Different object detection deep neural networks such as Region-based Convolutional Neural Networks (R-CNN) i.e. Fast R-CNN [5], Faster R-CNN [6] & Mask R-CNN [7], YOLO (You Only Look Once) [8] and SSD (Single Shot Detector) [9] network have been proposed to simultaneously classify and localize the objects in natural images.

Different approaches for brain tumor detection, classification and segmentation have been proposed. Ezhilarasi et al. [10] proposed a method that uses Faster R-CNN with the AlexNet model to detect and classify the brain tumor into multiple classes namely benign, malignant, glial and astrocytoma. Avsar et al. [11] and Bhanothu et al. [12] used Faster R-CNN algorithm to detect and classify the brain tumor into three different classes i.e. glioma, meningioma, and pituitary tumor. Hirarichal based tumor segmentation method was proposed by Sahaai et al. [13] which uses the SSD-MobileNetV2 for tumor detection and to classify the tumor images and non-tumor images. This process was then followed by an image thresholding technique for brain tumor segmentation. Kaldera et al. [14] employed Faster R-CNN to detect and classify the brain tumor MRI data in two classes namely meningioma and glioma tumor. For tumor segmentation, edge detection algorithms like Prewitt and Sobel operators were used to identify the tumor mask. Masood et al. [15] used Mask-RCNN for brain tumor detection and segmentation. Although the results of these analyses are encouraging but stratified splitting of data into training, validation and testing sets was not considered. In addition, a comparison with existing state-of-the-art detection and segmentation deep CNNs was rarely performed.

Object detection methods employ different algorithms to localize and classify objects e.g. Faster R-CNN [6], YOLO [8] and SSD [9]. The performance of these networks is often evaluated on natural images [16–18] using the COCO dataset or DOTA (Dataset of Object detection in Aerial images) dataset etc. However, the performance of detection networks is generally unclear in medical imaging data. So far, promising results have been shown in literature for brain tumor detection and segmentation, but very limited work has been done for integrating the detection, classification, and segmentation into a single module for evaluating models on multi-class MRI data of brain tumors. To the best of our knowledge, none of the previous studies has compared the classification and detection performance of deep learning-based detection networks for multiclass brain tumors.

This study aims to address simultaneous classification and segmentation problems by cascading existing deep learning-based object detection and segmentation algorithms. The classification and detection performance of three state-of-the-art deep learning-based object detection networks (Faster R-CNN, YOLO & SSD) was compared for three tumor types, namely meningioma, glioma and pituitary tumors. Furthermore, the best-performing detection network

was paired with 2D U-Net [19] or pixel-wise segmentation of abnormal tumor cells.

Materials and methods

Brain tumor dataset

In this study two publicly available brain tumor datasets were used: (i) Brain Tumor Figshare (BTF) dataset [20] and (ii) Brain Tumor Segmentation (BRATS) challenge 2018 dataset [21–23]. BTF dataset comprises of T1-weighted contrast enhanced (T1c-w) MR Images with three types of brain tumors: (i) meningioma, (ii) glioma and (iii) pituitary tumors. The repository provides a total of 3064 slices for 233 subjects (in-plane resolution = 512×512 , pixel size = 0.49×0.49 mm², slice thickness = 6 mm, slice gap = 1 mm) [4] and the corresponding tumor segmentation binary masks.

BRATS 2018 dataset contains 285 cases of glioma tumor: (i) 210 cases of high-grade glioma (HGG) and (ii) 75 cases of low-grade glioma (LGG). BRATS 2018 Dataset contains four MRI modalities for each case: T1-weighted (T1), T2-weighted (T2), T1c-w and fluid attenuated inversion recovery (FLAIR) images. The data was already preprocessed i.e. skull-stripped and resampled to $1 \times 1 \times 1$ mm isotropic resolution [24]. The corresponding tumor segmentation mask labels and the skull-stripped input images for each glioma case were of size $240 \times 240 \times 155$.

In this work, 2D data was used to evaluate the state-of-the-art detection networks like Faster R-CNN, YOLO, and SSD. These CNN are primarily designed to extract features from 2D inputs, aligning with their conventional architecture and purpose. Moreover, processing 2D data offers computational advantages, especially when considering limited computational resources.

Study design

This section describes the flow of the proposed method to perform brain tumor detection, classification and segmentation simultaneously using T1c-w MRI scans. Figure 1 summarizes the design of this study. Pre-processing steps applied to the BTF data include resizing, skull stripping and data labeling while for BRATS 2018 data, only resizing and data labeling were performed as the data was already skull stripped. Initially, the performance of three state-of-the-art deep learning-based object detection networks i.e. Faster R-CNN [5], YOLO version 4 (YOLOv4) [25], YOLO version 5 (YOLOv5) [26] and SSD [9] network was compared for brain tumor detection and classification using BTF dataset. These detection networks localize the brain tumor within a bounding box and classify the brain tumor into its types i.e.

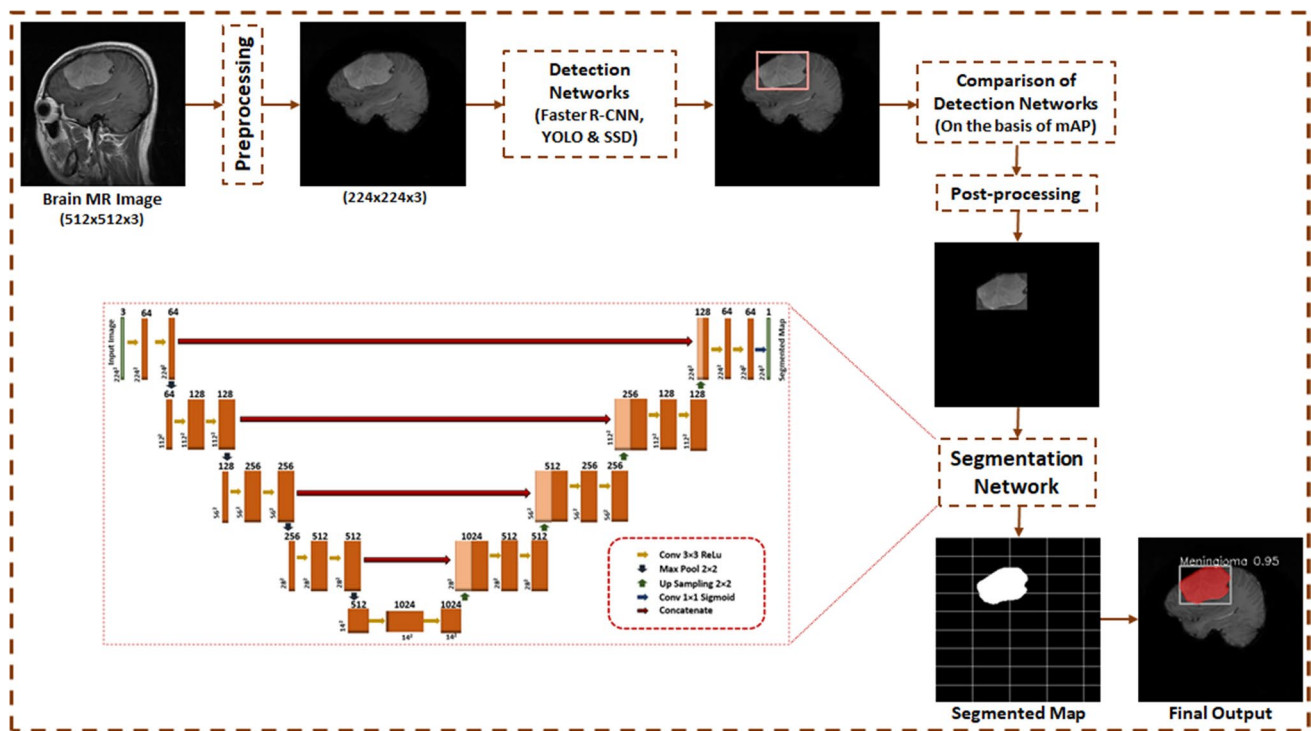


Fig. 1 The proposed methodology of brain tumor detection, classification, and segmentation; the best-performing detection network is paired with 2D U-Net for pixel-wise tumor segmentation

glioma, meningioma, and pituitary tumor for the BTF dataset and HGG and LGG for BRATS dataset. The performance of detection networks was evaluated on the basis of mean Average Precision (mAP). The best-performing detection network was further fine-tuned on BRATS 2018 challenge to detect and classify the grades of glioma tumor i.e. HGG and LGG. For pixel-wise tumor segmentation, the detected patches from the best-performing detection network were used. The performance of the segmentation network was evaluated on the basis of Dice Score Coefficient (DSC). The detection and segmentation performance of the proposed method was also compared with the existing state-of-the-art detection and segmentation network i.e. Mask R-CNN based on mAP and DSC.

All the networks were trained on training data and losses were optimized on the validation data together with hyperparameter tuning. Finally, the performance of each network was evaluated on the held-out test data.

Image preprocessing

In the image preprocessing step, firstly, T1c-w images and the corresponding segmentation masks from BTF and BRATS 2018 data were resized to 224×224 pixels. To provide ground truth bounding boxes for training and validating the detection networks, the binary mask of the segmented

region of both the datasets was localized by computing bounding box coordinates in x–y plane of each slice using a custom script written in Python. In the last step of data pre-processing, skull stripping of T1c-w images of BTF dataset was performed using FSL [27] – Brain Extraction Tool (BET) [28] to remove the non-brain tissues from MR Images.

Data splitting

All 233 cases of glioma, meningioma, and pituitary tumor from the BTF dataset acquired with T1c-w imaging were used for modeling. So, to maintain the uniformity and consistency among the datasets, only T1c-w image modality was chosen from BRATS 2018 dataset to evaluate the model performance. Also contrast enhancement of the tumor region in T1c-w images helps to better visualize the tumor tissues [29]. BRATS dataset contains 44,155 images (patient\slices: 285\155). Both the dataset was split into three sets i.e. train, validation and test using stratified splitting on the patient level; 70% of the data was used for training (patient\images: BTF = 168\2200, BRATS = 200\30922), 10% for validation (patient\images: BTF = 19\272, BRATS = 29\4417) and 20% of the data was held aside for final testing (patient\images: BTF = 46\592, BRATS = 57\8835) of the networks.

To mitigate overfitting, we performed three-fold cross-validation [30] on both the BTF and BRATS 2018 datasets. The BTF dataset comprises 3064 images. A three-fold cross-validation approach was applied, dividing the dataset into three folds: fold 1 (k1) with 1022 images, fold 2 (k2) with 1022 images, and fold 3 (k3) with 1020 images. Each fold served as the test set once, while the remaining folds were used for training and validation.

The BRATS 2018 dataset consists of 285 subjects, with a total of 44,175 images. Like the BTF dataset, three-fold cross-validation was applied. Here, each fold contained approximately one-third of the subjects, with the other two-thirds used for training and validation. This procedure was repeated three times, with each fold being utilized once as the test set. The results from each fold were averaged to determine the final mAP and DSC scores.

By using this cross-validation methodology, we ensured robust performance evaluation and minimized overfitting in our proposed method.

Detection and segmentation network

We investigated the performance of three state-of-the-art deep learning-based object detection networks (Faster R-CNN, YOLO & SSD) and compared their performances for brain tumor detection and classification. Each network was trained on training data and the internal validation was performed on the validation data. Finally, the network with the best performance was applied to test the data for final predictions.

Firstly, for the BTF dataset, detection networks including Faster R-CNN, YOLOv4, YOLOv5 and SSD were trained on the training dataset for the detection and classification of brain tumors into three classes i.e. meningioma, glioma, and pituitary tumor. The Faster R-CNN was trained individually with three different backbones for feature extraction i.e. VGG16 [31], ResNet-50 [32] and DenseNet [33], while the SSD network was trained with VGG16 backbone. Each network was tuned over a range of parameters (learning rate = [0.0001–0.01], number of epochs = [50–200], optimizer = [Adam, SGD]) to find the best set of hyperparameters that provide maximum results on the validation data.

As an output detection network localizes a tumor region within a bounding box and returns a confidence score on a scale of 0–1 for each T1c-w MR image. This score represents how likely the bounding box contains the tumor of interest. The overall classification and detection performance of each detection network was evaluated based on mAP.

In the next step, the detected tumor regions from the best-performing detection network were extracted to segment the brain tumor at the pixel level.

For pixel-wise tumor segmentation, we employed a 2D U-Net architecture [19]. The detected bounding boxes on

MR images from the best-performing detection network were cropped and converted to a fixed-size (i.e. 224×224) using zero padding for both the training and validation data. In the case of multiple detections in a single slice, only one bounding box with the highest confidence score was extracted for tumor segmentation. This strategy aimed to mitigate false positives and prioritize areas estimated as more likely to contain tumors by the network. These patches were then fed into the segmentation network. 2D U-Net was trained on training patches generated from training and losses were optimized on the validation patches generated from the validation data. The segmentation network was tuned over a range of hyperparameters i.e. learning rate = [0.0001–0.01], number of epochs = [60–100], optimizer = [Adam, SGD, RMSprop] to find the best set of hyperparameters that provide maximum results on the validation data. Finally, the segmentation network, with the best set of hyperparameters was applied to the test data for final predictions.

In a separate experiment, the best-performing detection network was further fine-tuned for 70 epochs with a learning rate of 0.001 on the BRATS 2018 dataset to detect and classify HGG and LGG. The purpose of fine-tuning was to specifically recognize the grades of glioma tumor along with the different classes of brain tumor. In our experiment, fine-tuning was carried out using the pre-trained weights from the BTF dataset to detect HGG and LGG from BRATS dataset.

Comparative analysis with standard 2D U-net and mask RCNN

Standard 2D U-Net architecture was also trained on the whole training data with model hyperparameter tuning based on the validation data without extracting the detected patches for a comparative analysis. The segmentation network was tuned over a range of hyperparameters i.e. learning rate = [0.0001–0.01], number of epochs = [60–100], optimizer = [Adam, SGD, RMSprop] to find the best set of hyperparameters that provide maximum results on the validation data. Furthermore, the detection and segmentation quality of our proposed method was also compared with an already available joint detection and segmentation network i.e. Mask R-CNN. We trained the Mask-RCNN model with ResNet-101 backbone on the training set of BTF data and internal validation was performed on the validation data. Finally, the model with the best set of hyperparameters was applied to the test data for final predictions.

All models were trained using Python 3.11 with Keras and TensorFlow on an AMD Ryzen 7 5800H CPU with a clock frequency of 3.19 GHz, 24 GB RAM, and an NVIDIA GeForce GTX 3070 GPU.

Evaluation parameters

Detection networks were evaluated on the basis of mAP that was computed by taking the mean of Average Precision (AP) for all the classes present in the dataset. It compares the ground-truth bounding box to the detected box and returns a score [34].

$$mAP = \frac{1}{n} \sum_{k=1}^n AP_k \quad (1)$$

The performance of the segmentation network was evaluated on the basis of DSC that provides the overlapping pixels between the ground truth (A) and the segmented result (B) [35].

$$DSC(A, B) = \frac{2|A \cap B|}{|A| + |B|} \quad (2)$$

The mAP metric measures the precision across different recall values that evaluates the detection network's performance. Meanwhile, the DSC quantifies the overlap between predicted and ground truth segmentation masks, indicating the segmentation accuracy. So, the combination of mAP and DSC offers a balanced evaluation strategy for both detection and segmentation network performance, respectively.

Results

Table 1 shows the training, validation, and testing results of different detection networks for BTF and BRATS 2018 datasets. For the BTF dataset, the results of different detection networks i.e. Faster R-CNN, YOLOv4, YOLOv5 and SSD were compared for meningioma, glioma, and pituitary tumor and for BRATS 2018 dataset, the performance of YOLOv5 was evaluated to detect HGG and LGG. For the BTF dataset, YOLOv5 performed best in validation with the final set of hyperparameters (i.e. mAP: YOLOv5 = 86.3%, YOLOv4 = 73.9%, Faster R-CNN = 67%, SSD = 67.8%). YOLOv5 also achieved the highest mAP when tested on

the unseen data with a learning rate of 0.01; 200 epochs, SGD optimizer and performed better than other detection networks (e.g. mAP: YOLOv5 = 89.5%, YOLOv4 = 85.9%, Faster R-CNN = 68%, SSD = 74.2%). Furthermore, the performance of YOLOv5 with and without fine-tuning was evaluated on BRATS 2018 dataset to detect HGG and LGG on the basis of mAP. YOLOv5 with fine-tuning performed better on validation data (e.g. mAP: YOLOv5 (with fine-tuning) = 85.3%, YOLOv5 (without fine-tuning) = 80.7%) with the best set of hyperparameters i.e. learning rate 0.001; epochs 70 and Adam optimizer and achieved good results on the test data (i.e. mAP: YOLOv5 (with fine-tuning) = 81%, YOLOv5 (without fine-tuning) = 69.3%).

Figure 2 shows the output results of the detection algorithms when applied to the sample images of the BTF dataset. From the figure, we demonstrate that YOLOv5 network accurately detected the tumor class with a high confidence score and location of the tumor. In contrast, the other detection networks i.e. Faster R-CNN, YOLOv4 and SSD did not perform the precise localization of brain tumor when compared with the ground truth labels.

Table 2 shows the segmentation results of the proposed and direct segmentation network in terms of DSC on the BTF dataset. Proposed method (YOLOv5 + 2D U-Net) performed best in validation with the final set of hyperparameters using learning rate of 0.001; 60 epochs and Adam optimizer (i.e. DSC: proposed method = 90.4%, direct segmentation method = 89.6%). The proposed method also achieved the highest DSC on the test data and performed better than the standard 2D U-Net (i.e. DSC: proposed method = 88.1%, direct segmentation method = 80.5%).

The segmented tumor regions by both segmentation methods are shown in Fig. 3. The proposed method performed precise segmentation of brain tumor using the extracted patches from YOLOv5 network. However, false detection was observed using standard 2D U-Net for tumor segmentation when compared with the ground truth label and thus resulted in reduced DSC.

Table 3 presents the results of Mask R-CNN on the test data of the BTF dataset in terms of mAP and DSC. Mask R-CNN

Table 1 Comparison of object detection networks (Faster R-CNN, YOLO (v4 & v5) and SSD) on BTF dataset and the results of YOLOv5 model on BRATS 2018 dataset with and without fine-tuning on the basis of mAP (%)

Dataset	Method	Learning Rate	Optimizer	Batch Size	Epochs	mAP		
						Train data	Validation data	Test data
BTF	Faster R-CNN	0.0001	SGD	1	60	71%	67%	68%
	YOLOv4	0.01	SGD	64	100	89.3%	73.9%	85.9%
	YOLOv5	0.01	SGD	8	200	99%	86.3%	89.5%
	SSD	0.0001	Adam	32	150	90.7%	67.8%	74.2%
BRATS 2018	YOLOv5 (with fine-tuning)	0.001	Adam	8	70	90.9%	85.3%	81%
	YOLOv5 (without fine-tuning)	0.01	SGD	8	100	87.4%	80.7%	69.3%

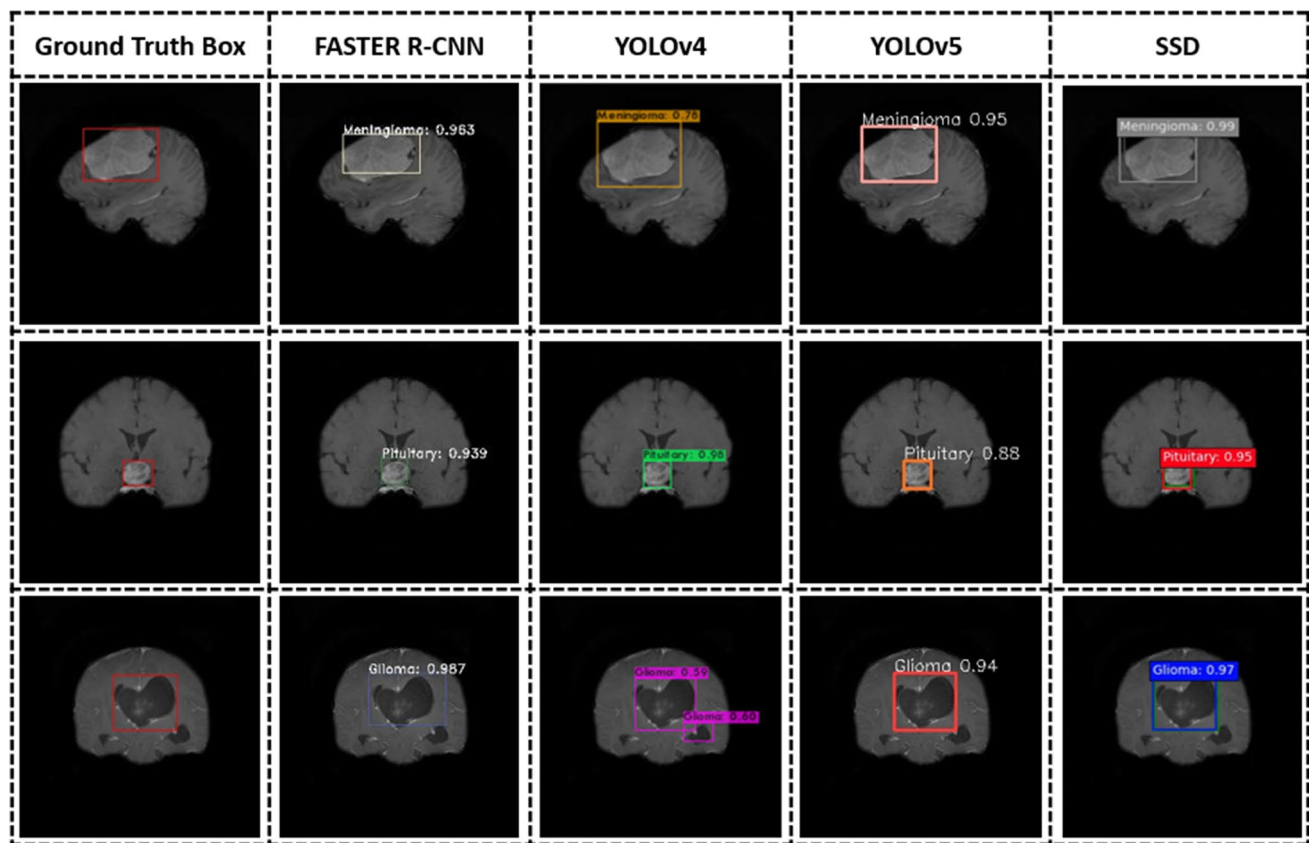


Fig. 2 Comparison showing the detected meningioma, pituitary, and glioma tumor from Faster R-CNN, YOLOv4, YOLOv5 and SSD model on BTF dataset. YOLOv5 outperformed the other detection models by detecting the large and small-sized tumor accurately

Table 2 Comparison of the proposed method with the direct segmentation method (2D U-Net) on BTF dataset in terms of dice score coefficient (%)

Method	Learning rate	Batch size	Epochs	Train data	Validation data	Test data
DSC						
Proposed method (YOLOv5 with 2D U-Net)	0.001	32	60	96.4%	90.4%	88.1%
Direct segmentation	0.001	32	60	90.5%	89.6%	80.5%

achieved an mAP of 67% and a DSC of 44.2% on the test data with a learning rate of 0.001, 60 epochs and SGD optimizer. Comparison of the predicted results obtained from the proposed method and Mask R-CNN is shown in Fig. 4. From this figure, we demonstrate that Mask R-CNN segmented the normal cells along with abnormal tumor cells as compared to ground truth labels, while the proposed method performed accurate pixel-wise segmentation of the abnormal tumor cells.

Discussion and conclusion

In this work, a comparison of deep learning-based object detection networks was performed for simultaneous classification and detection of glioma, meningioma, and pituitary tumors using BTF data and LGG and HGG using the BRATS dataset. Among all the trained detection networks

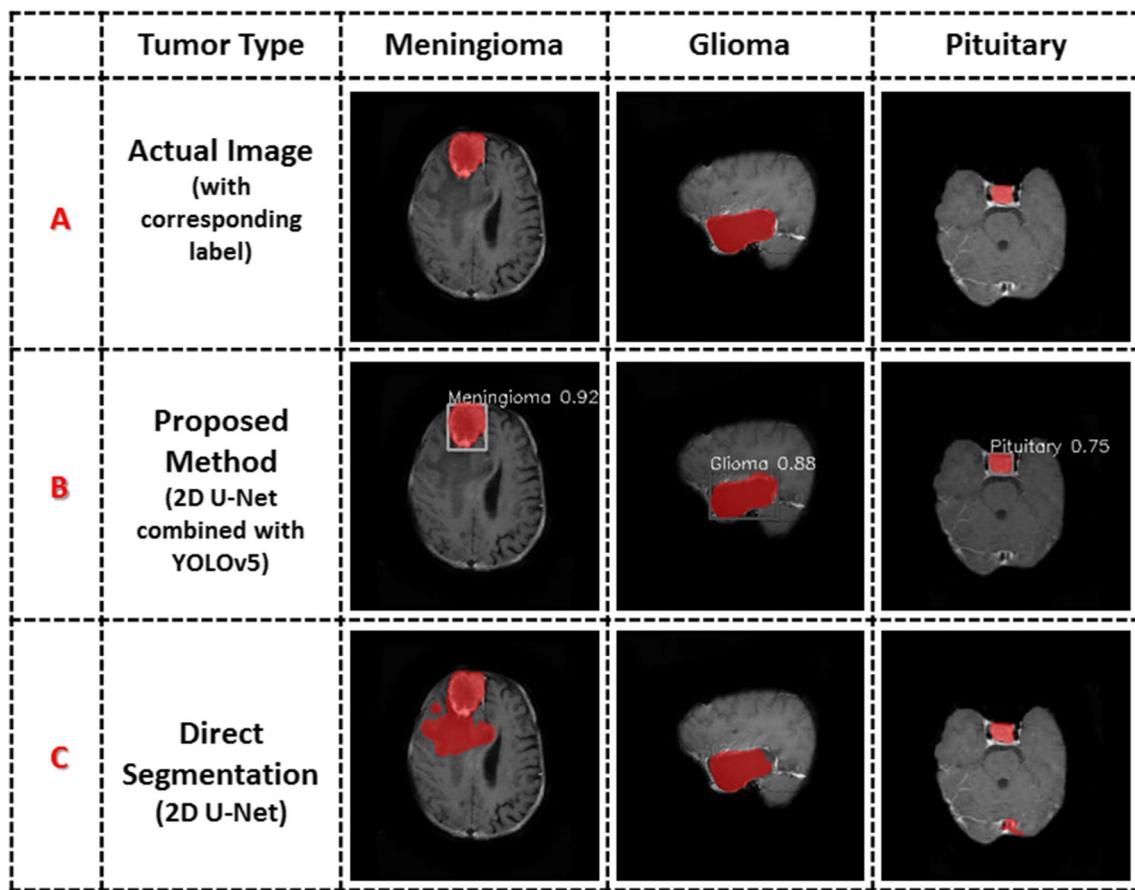


Fig. 3 Comparison of the segmentation results obtained from direct segmentation method (2D U-Net) and the proposed method (YOLOv5 with 2DU-Net) on BTF dataset: Brain images with actual tumor against the rest of the images (A), segmented tumor region using the proposed method (B), segmented tumor region using 2D

U-Net (i.e. direct segmentation method) (C). The proposed method performs accurate pixel-wise tumor segmentation using the detected region of interest from YOLOv5 model as compared to the direct segmentation method

Table 3 Results of Mask R-CNN on BTF dataset with reference to mAP (%) and dice score coefficient (%)

Method	Learning rate	Optimizer	Batch size	Epochs	Test data	
					mAP	DSC
Mask R-CNN	0.001	SGD	2	60	67%	44.2%

(i.e. Faster R-CNN, YOLOv4, YOLOv5 and SSD), YOLOv5 achieved the highest results for tumor detection and classification on the BTF and BRATS datasets (i.e. mAP: BTF = 89.5%, BRATS = 81.32%).

Furthermore, combining YOLOv5 with 2D U-Net achieved the highest results for tumor segmentation (DSC = 88.1%) compared to the standard 2D U-Net and Mask R-CNN for BTF data.

From the previous related work, we could not conclude about the object detection algorithm that works best for brain tumor detection. Therefore, we selected three state-of-the-art deep learning-based detection networks i.e. Faster R-CNN, YOLO (version 4 & 5) and SSD to investigate which

algorithm performs best for the detection and classification of different types of brain tumors. Different approaches have been proposed in the literature for brain tumor detection and classification e.g. Avsar et al. [11] (classification accuracy = 91.6%) and Bhanothu et al. [12] (mAP = 77.6%) used Faster R-CNN algorithm to detect and classify the brain tumor into three different classes; however, brain tumor segmentation was not performed. Sahaai et al. [13] used SSD-MobileNetV2 for tumor detection and achieved an overall accuracy of 90%. Only a few studies exist that performed tumor detection, classification, and segmentation simultaneously e.g. Masood et al. [15] (mAP = 94% & DSC = 95%) employed Mask R-CNN for brain tumor

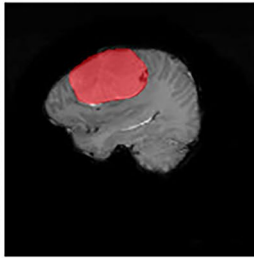
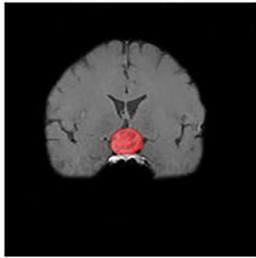
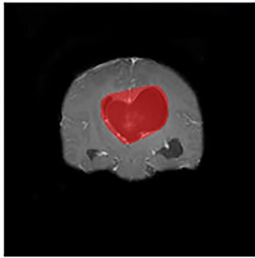
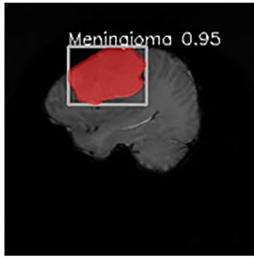
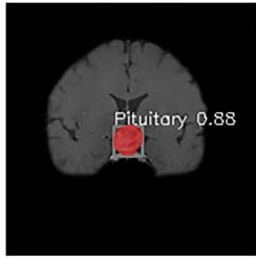
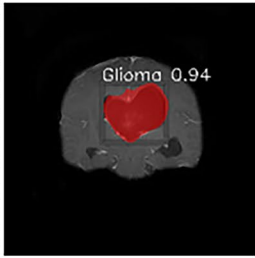
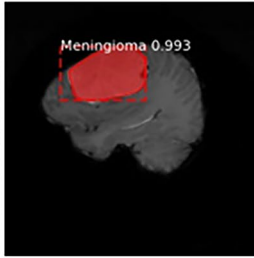
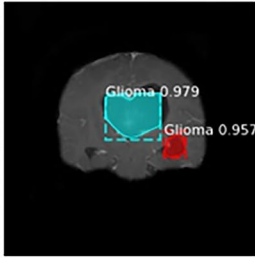
	Tumor Type	Meningioma	Pituitary	Glioma
A	Actual Image (with corresponding label)			
B	Proposed Method (2D U-Net combined with YOLOv5)			
C	Mask R-CNN			

Fig. 4 Comparison of the segmentation results obtained from Mask R-CNN and the proposed method (YOLOv5 with 2DU-Net) on BTF dataset: Brain images with actual tumor against the rest of the images (A), segmented tumor region using the proposed method (B),

segmented tumor region using Mask R-CNN (C). Proposed method performs accurate pixel-wise tumor segmentation using the detected region of interest from YOLOv5 model as compared to the existing detection & segmentation network i.e. Mask R-CNN

detection, classification and segmentation. In these papers, data was split randomly; stratified splitting was not considered which resulted in an inaccurate representation of data in each group [36] (data leakage problem). In this paper, stratified splitting was used for data splitting and our method achieved better detection and segmentation results ($mAP=89.5\%$, $DSC=88.1\%$) compared to the existing state-of-the-art detection and segmentation network i.e. Mask R-CNN ($mAP=67\%$, $DSC=44.2\%$) on BTF dataset.

Further, all the experiments in this paper were performed on T1c-w image modality as it enhances the contrast of the soft tissues, provides increased image details, and thus helps to better visualize the tumor tissues [29]. Furthermore, an important pre-processing in our work was the distribution of the dataset in each section (i.e. training, validation, and testing) for which stratified splitting was performed for both the datasets (BTF & BRATS). The purpose of stratified splitting was to preserve the same proportion of each class with respect to the number of patients in the whole dataset and to avoid data leakage problem.

A limitation of the current work is a limited sample size with only 3 tumor classes i.e. meningioma, glioma, and pituitary tumor. To overcome this problem, BRATS 2018 data was used to increase the number of classes and to detect the grades of glioma tumor as HGG and LGG. Furthermore, the proposed method requires two models (YOLOv5 & 2D U-Net) to train individually to perform brain tumor detection and segmentation compared to the existing detection and segmentation network i.e. Mask R-CNN. However, the results show that the proposed model provides better detection and segmentation results in terms of mAP and DSC as compared to the Mask R-CNN.

Another limitation is related to the selection of the bounding box with the highest confidence score in cases where multiple bounding boxes are detected. This approach may lead to the potential loss of parts of tumors if they span multiple bounding boxes. This is particularly concerning for tumors that do not fit well within a single bounding box. Future work will explore methods such as bounding box merging and uncertainty handling to improve segmentation

accuracy and ensure that the entire tumor is accurately captured.

In conclusion, we performed multi-class tumor detection, classification and segmentation simultaneously by combining deep learning-based detection and segmentation networks i.e. YOLOv5 and 2D U-Net. The results showed that the proposed method not only detects different types of brain tumors accurately but also delineates the tumor region precisely within the detected bounding box.

In the future, we will extend our analysis to incorporate updated versions of the BRATS dataset for a more comprehensive assessment. We will also integrate some extended versions of U-Net in our proposed method.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10334-024-01203-5>.

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Data availability In this study two publicly available brain tumor datasets were used: (i) Brain Tumor Figshare (BTF) dataset [20] and (ii) Brain Tumor Segmentation (BRATS) challenge 2018 dataset [21–23].

Declarations

Conflict of interest Rafia Ahsan declares that he has no conflict of interest. Iram Shahzadi declares that he has no conflict of interest. Faisal Najeeb declares that he has no conflict of interest. Hammad Omer declares that he has no conflict of interest.

Ethical approval All the procedures performed in the studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed consent Informed consent was obtained from all the individual participants included in the study before data acquisition.

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